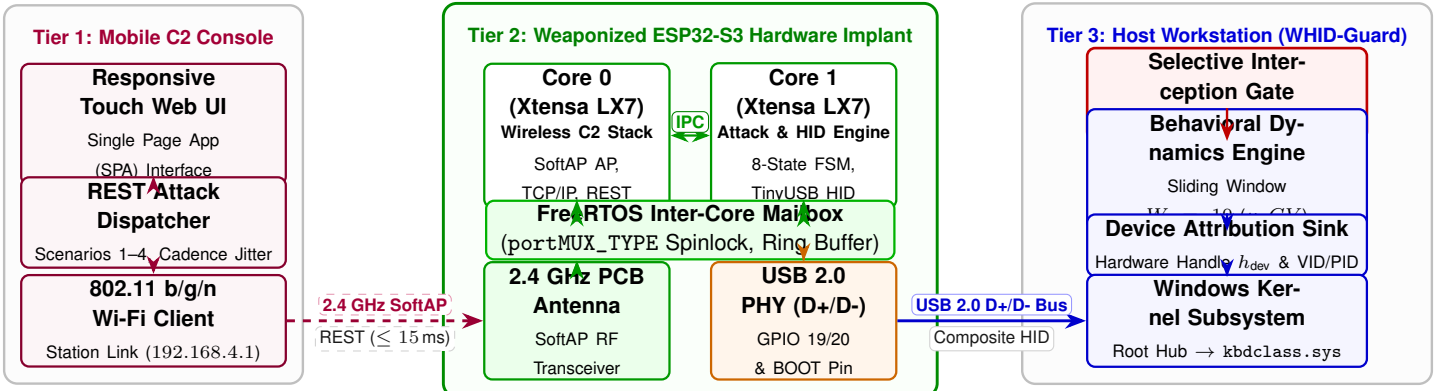

Tier 1: Mobile C2 Console
Responsive Touch Web UI
Single Page App (SPA) Interface
REST Attack Dispatcher
Scenarios 1–4, Cadence Jitter
802.11 b/g/n Wi-Fi Client
Station Link (192.168.4.1)
2.4 GHz SoftAP
REST (≤ 15 ms)
Tier 2: Weaponized ESP32-S3 Hardware Implant
Core 0 (Xtensa LX7)
Wireless C2 Stack
SoftAP AP, TCP/IP, REST
IPC
Core 1 (Xtensa LX7)
Attack & HID Engine
8-State FSM, TinyUSB HID
FreeRTOS Inter-Core Mailbox
(portMUX_TYPE Spinlock, Ring Buffer)
2.4 GHz PCB Antenna
SoftAP RF Transceiver
USB 2.0 PHY (D+/D-)
GPIO 19/20 & BOOT Pin
USB 2.0 D+/D- Bus
Composite HID
Tier 3: Host Workstation (WHID-Guard)
Selective Interception Gate
Behavioral Dynamics Engine
Sliding Window
Device Attribution Sink
Hardware Handle h_{dev} & VID/PID
Windows Kernel Subsystem
Root Hub → kbdclass.sys